

DeepCSAT: E-Commerce Customer Satisfaction Score Prediction

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GitHub link - <https://github.com/akshat-mishra-ds/E-Commerce-Customer-Satisfaction-Score-Prediction>

Google Drive link -

<https://drive.google.com/drive/folders/1mv0vKWdcLoAbcV0m4AUvP4LD4jtcEoNY?usp=sharing>

This project focuses on predicting Customer Satisfaction (CSAT) scores using Artificial Neural Networks (ANN) for an e-commerce platform. The objective is to leverage customer support interaction data to proactively estimate satisfaction levels and support business decision-making. The project includes data cleaning, exploratory data analysis, hypothesis testing, feature engineering, deep learning model development, evaluation, and local deployment using Streamlit.

Introduction

Customer satisfaction is a critical performance indicator in e-commerce. Understanding customer experience helps organizations improve retention, loyalty, and operational efficiency. Traditional survey-based measurement is reactive, whereas predictive analytics enables proactive intervention. This project develops a predictive system that estimates CSAT scores from support interaction data.

Problem Statement

The goal is to build a predictive model capable of estimating CSAT scores based on customer support interactions, product information, order attributes, and operational features.

Business Objectives

Improve customer experience, reduce dissatisfaction, optimize support operations, identify service bottlenecks, and support data-driven decision-making.

Dataset Description

The dataset contains customer support records including support channel, category, sub-category, customer remarks, order details, product category, item price, handling time, support hierarchy, tenure bucket, shift information, and CSAT score.

Methodology

The project followed a structured workflow: Data Understanding, Data Wrangling, Exploratory Data Analysis, Hypothesis Testing, Feature Engineering, Data Preprocessing, ANN Modeling, Evaluation, and Deployment.

Data Understanding

Dataset dimensions, feature types, missing values, duplicates, and target variable distribution were analyzed. CSAT score was identified as a multiclass classification target.

Data Wrangling

Missing values were investigated, duplicate records were analyzed, identifier columns were identified, and feature relevance was evaluated before preprocessing.

Exploratory Data Analysis

EDA included analysis of CSAT distribution, support channels, issue categories, product categories, agent tenure, shifts, handling time, item prices, and feature relationships. Insights were documented with business impact statements.

Hypothesis Testing

Statistical tests were performed to evaluate relationships between support channels, product categories, agent tenure, and customer satisfaction outcomes.

Feature Engineering

Datetime variables were transformed into business-relevant features such as Response_Time_Hours, Survey_Delay_Hours, Order_Age_Hours, Order_Hour, Order_Day, and Order_Month. Missing value indicators were also created to preserve information. Additional remark-length and operational features were engineered.

Data Preprocessing

Identifier columns were removed. Missing values were handled using suitable imputation strategies. Categorical variables were encoded using one-hot encoding. Numerical features

were standardized using StandardScaler. Train-test split was performed to prepare data for modeling.

Artificial Neural Network Model

A deep learning ANN architecture was implemented using TensorFlow and Keras. The model included Dense layers, Batch Normalization, Dropout regularization, Early Stopping, and ReduceLROnPlateau callbacks. The objective was to learn nonlinear relationships among customer support variables.

Model Architecture

Input Layer → Dense(128) → Batch Normalization → Dropout(0.20) → Dense(64) → Batch Normalization → Dropout(0.20) → Dense(32) → Output Layer (Softmax). Adam optimizer and sparse categorical cross-entropy loss were used.

Model Evaluation

The final model achieved approximately 70.04% Accuracy, 55.70% Precision, 70.00% Recall, and 59.93% F1 Score. Training and validation curves demonstrated stable learning behavior with limited overfitting.

Results and Interpretation

The ANN successfully learned useful patterns from customer support data. The results indicate that customer satisfaction can be reasonably predicted using historical interactions. The model provides actionable insights for service quality improvement.

Business Insights

Response time, support channel, issue category, product category, and operational attributes influence customer satisfaction. Organizations can use these insights to optimize support workflows and improve customer experience.

Local Deployment

The trained ANN model, scaler, and feature structure were exported from Google Colab and deployed locally using Streamlit in VS Code. The application accepts user inputs and generates real-time CSAT predictions.

Streamlit Deployment Screenshot

localhost:8501

DeepCSAT Prediction App

Predict Customer Satisfaction Score

Item Price

1500.00 - +

Response Time (Hours)

2.00 - +

Order Age (Hours)

100.00 - +

Remark Length

50 - +

Predict CSAT

Predicted CSAT Score: 5

Limitations

The dataset contains significant missing values and class imbalance. The model tends to favor the majority class. Future balancing strategies can improve minority-class prediction performance.

Future Work

Future improvements include SMOTE-based balancing, advanced NLP for customer remarks, SHAP explainability, hyperparameter optimization, XGBoost comparison, and cloud deployment.

Conclusion

The project successfully developed an ANN-based customer satisfaction prediction system. The solution demonstrates the practical application of deep learning in customer experience analytics and provides a foundation for proactive service optimization in e-commerce environments.

References

TensorFlow Documentation, Scikit-Learn Documentation, Streamlit Documentation, AlmaBetter Deep Learning Curriculum, and relevant customer satisfaction analytics literature.